



Datasheet

OpenTherm Gateway

Part No.: OpenTherm Gateway-R1

Manufacturer: ISYSTEMS AUTOMATION S.R.L.

Hardware Version: V1.0

Architecture: Modular (MCU Board + Relay Board)

Controller: ESP32-WROOM-32U-N16



Functional Overview

The OpenTherm Gateway is a compact embedded automation controller designed for integration with OpenTherm heating systems and environmental sensors. The system is built around the ESP32 platform and provides reliable communication with heating equipment while enabling local control logic and automation.

The Gateway supports multiple power input options (24V DC or widerange AC/DC) and integrates an OpenTherm interface for communication with compatible boilers and HVAC devices. Additional capabilities include a relay output for actuator control, 1-Wire temperature sensor support, USB connectivity for firmware programming, and Wi-Fi communication through the ESP32 module.

The system is implemented using two hardware boards:

Board	Function
MCU Board	CPU, USB interface, OpenTherm interface, logic
Relay Board	Power supply, relays, 1-Wire diodes

Technical Specifications

Specification	Details
Microcontroller	ESP32-WROOM-32U-N16 (dual-core)
Power Input	24 V DC nominal or 85–265 V AC, 47–63 Hz (L / N)
RTC	RTC (PCF8563); backup battery holder, battery not fitted
Document revision	DS-OpenThermGateway Rev. B · 2026-09 · Hardware V1.0
Relay outputs	1 × relay, C and NC only (SPST-NC); 3 A @ 250 VAC per contact (module limit)
Temperature Inputs	2 × 1-Wire (DS18B20 compatible)
User Interface	1 button, 4 LEDs (PWR, 1 user, 1 status, relay state)
OpenTherm Interface	1 × OpenTherm bus (OT+ / OT-) with optocoupler isolation
Wi-Fi	Wi-Fi (ESP32)
USB	USB-C (ESD protected, CC detection, data to ESP32)
Typical Power Consumption	3 W
Operating temperature	0 °C to +40 °C
Storage temperature	–10 °C to +55 °C
Relative humidity	0–90 % RH, non-condensing
Ingress protection	IP20 (inside cabinet only)
Installation	Indoor control cabinet only; not for outdoor or exposed installation
Maximum altitude	2000 m
Pollution degree	2
Dimensions	35.5 × 90.6 × 67.3 mm (L × W × H)
DIN width	2 modules (≈ 35.5 mm)
Mounting	35 mm DIN rail
Enclosure	PC/ABS industrial enclosure
Terminal type	Pluggable screw terminal blocks, 5.08 mm pitch
Wire cross-section	0.2–2.5 mm ² (AWG 24–12)
Tightening torque	0.4–0.6 Nm
Net weight	TBD
Gross weight	TBD
Pack size	140 × 96 × 95 mm (L × W × H)

Top Terminal Block (1–6)

Pos	Marking	Signal	Description
1	Gnd	ONEWIRE	1-Wire ground
2	D1	ONEWIRE	1-Wire bus 1 data
3	D2	ONEWIRE	1-Wire bus 2 data
4	+5V	ONEWIRE	1-Wire sensor supply, max 50 mA
5	O+	OT	OpenTherm bus positive

6	O-	OT	OpenTherm bus negative
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Bottom Terminal Block (1–6)

Pos	Marking	Signal	Description
1	0V	POWER_DC	24 V DC return
2	+V	POWER_DC	24 V DC input
3	L	POWER_AC	AC line 85-265 V (MAINS)
4	N	POWER_AC	AC neutral (MAINS)
5	C	RELAY	Relay common
6	NC	RELAY	Relay normally closed

*Relay outputs are **not internally fused**. External overcurrent protection must be provided in accordance with applicable standards and installation requirements. Use appropriately rated fuses or circuit breakers and consider external contactors for loads exceeding 3 A or for inductive and high inrush loads.

Installation, Environmental & Mechanical

Category	Specification	Details
Terminal Specifications	Terminal type	Pluggable screw terminal blocks, 5.08 mm pitch
	Terminal pitch	5.08 mm
	Wire cross-section	0.2–2.5 mm ² (AWG 24–12)
	Conductor type	Solid or stranded copper
	Stranded wire	Ferrules recommended
	Tightening torque	0.4 Nm (max)
Environmental Ratings	Operating temperature	0 °C to +40 °C
	Storage temperature	-10 °C to +55 °C
	Relative humidity	0–90 % RH, non-condensing
	Ingress protection	IP20 (inside cabinet)
	Installation	Indoor control cabinet only; not for outdoor or exposed installation
	Terminal protection	All wiring terminals must be protected against accidental contact
Mechanical & Packaging	Product dimensions	35.5 × 90.6 × 67.3 mm (L × W × H)
	DIN units	2 modules (≈ 35.5 mm)
	Mounting	35 mm DIN rail
	Net weight	TBD
	Gross weight	TBD
	Pack size	140 × 96 × 95 mm (L × W × H)
	Mechanical drawing	Mechanical drawing: front and side view, dimensions in mm

Cable Recommendations & Shield Grounding

This section applies to **Temperature (1-Wire)**, and **OT**. Use shielded, twisted constructions and bond shields correctly to reduce EMI and ground-loop issues.

General Routing Rules

- Route low-level signal cables (1-Wire/OT) separately from mains, relay outputs, contactors, VFD motor cables, and power wiring.
- If crossing power cables is unavoidable, cross at **90°**.
- Keep cable runs as short as practical and avoid parallel runs with high-current conductors.

Temperature Cable

1-Wire (DS18B20)

- **Recommended:** shielded 3-core (+5V / DATA / GND) for typical installations
- **High-EMI / long runs:** shielded pairs + overall shield (e.g. **LI2YCY PIMF 2×2×0.50**)
- **Topology:** daisy-chain (bus). Avoid star wiring.
- **Stubs:** keep sensor stubs ≤ 0.5 m.
- **Pull-up (DATA):** **4.7 k Ω** typical; **2.2–3.3 k Ω** for long/heavy loads.

OT Cable

- **Construction:** twisted pair
- **Shielding:** overall shield is recommended in cabinets; use individually shielded pairs + overall shield in high-EMI
- **Examples:** **J-Y(ST)Y 2×2×0.5 mm²** or **LI2YCY PiMF 2×2×0.50**

Shield Grounding

- **Default recommendation:** bond cable shield(s) to cabinet **PE/EMC ground at the PLC end only**.
- **Do not** connect shields to signal terminals (1-Wire/OT).
- If both ends are in equipotential bonded cabinets, shields may be bonded at both ends using proper 360° clamps.

Compliance & Certifications

The HomeMaster OpenTherm Gateway is CE marked and designed to comply with the applicable European Union directives. The manufacturer, ISYSTEMS AUTOMATION (HomeMaster brand), maintains the technical documentation and a signed EU Declaration of Conformity (DoC).

EU Directives

EMC 2014/30/EU · LVD 2014/35/EU · RED 2014/53/EU · RoHS 2011/65/EU

Harmonised Standards

Area	Standards
EMC	EN 61000-6-1 (Immunity) · EN 61000-6-3 (Emissions)
Electrical Safety	EN 62368-1
Radio	EN 300 328 · EN 301 489-1 · EN 301 489-17
RoHS	EN IEC 63000

Radio

The product integrates a pre-certified ESP32 Wi-Fi radio module (2.4 GHz). Final product conformity with the Radio Equipment Directive is demonstrated by the maintained technical documentation and conformity assessment of the complete device.

Safety notice

cross-mains use would require reinforced insulation that the current PCB revision (V1.0) does not provide. Connect C/NC only to the same mains supply that feeds L/N, or to SELV/LPS circuits.

current PCB revision (V1.0) provides basic insulation only; connect C/NC only to the same mains supply that feeds L/N, or to SELV/LPS circuits

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